

Secure & Scalable Static Website Hosting on AWS

Instructions: In today's digital landscape, businesses and individuals require reliable, secure, and globally accessible web platforms to establish their online presence. Traditional hosting solutions often involve complex infrastructure management and higher costs, making them less efficient for static websites.

This project, **"Secure & Scalable Static Website Hosting on AWS"**, addresses these challenges by leveraging Amazon Web Services to deliver a cost-effective, highly available, and secure hosting solution. Using **Amazon S3, Route 53, AWS Certificate Manager, and Amazon CloudFront**, the system provides end-to-end infrastructure for hosting static content with custom domain integration, DNS management, and global content delivery.

By implementing SSL/TLS certificates and CloudFront caching policies, the project ensures secure communication, compliance with modern standards, and optimized performance across regions. The result is a professional-grade static website that demonstrates practical skills in cloud infrastructure setup, domain management, and performance optimization.

The project demonstrates:

- Import a domain name on AWS
- Create a hosted zone and manage its DNS records;
- Create S3 Buckets to host website pages;
- Configure Route 53 to link the domain name to the website;
- Create an SSL certificate for the website using AWS Certificate Manager (ACM);
- Create CloudFront distributions to secure the website in HTTPS.

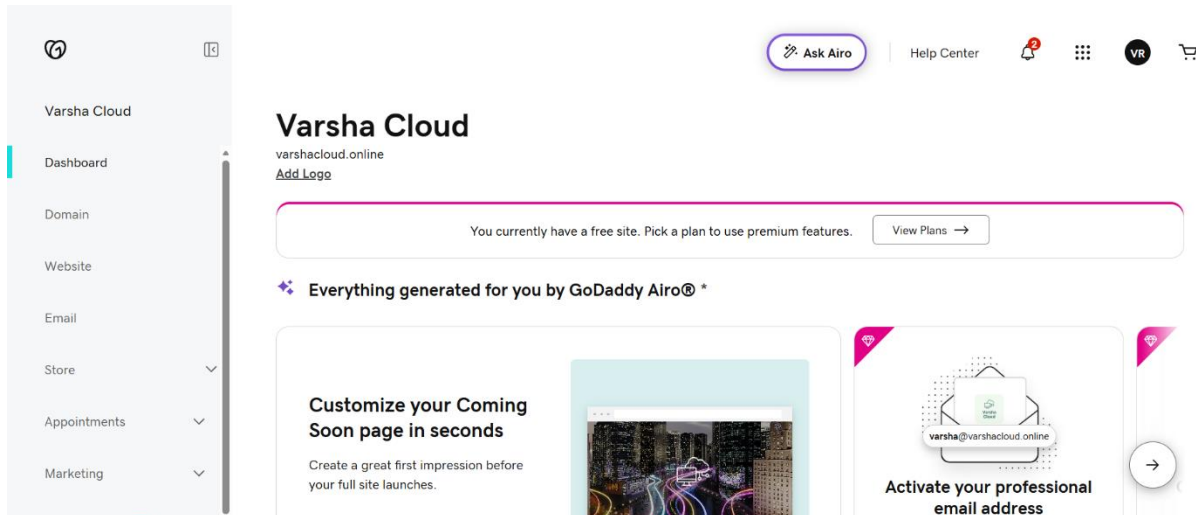
Tools & Technologies Used:

- **Amazon S3** – Static website hosting and storage.
- **Amazon Route 53** – DNS management and domain integration.
- **AWS Certificate Manager (ACM)** – SSL/TLS certificate generation.

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- **Amazon CloudFront** – Content delivery network (CDN) for global distribution.

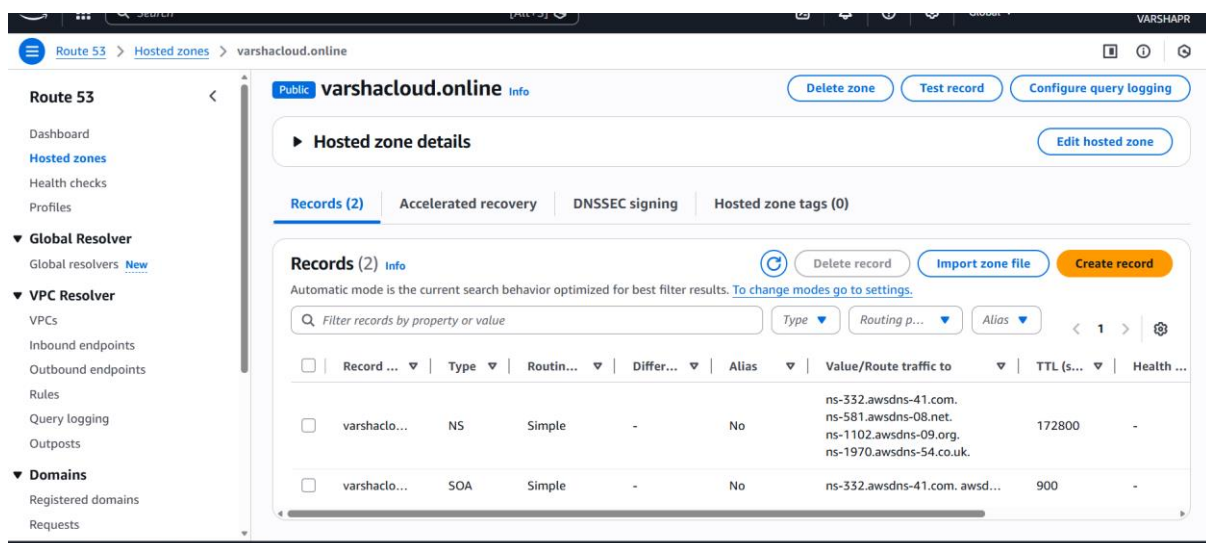
Use a domain name:



- Registered a custom domain to establish a professional online presence.
- Ensured domain availability and ownership through a registrar.

Create hosted Zone:

First, create your hosted zone, where you specify your domain name, in the example *cif-project.com*; then specify that you want your hosted zone to be public, otherwise it will be inaccessible from the public internet:



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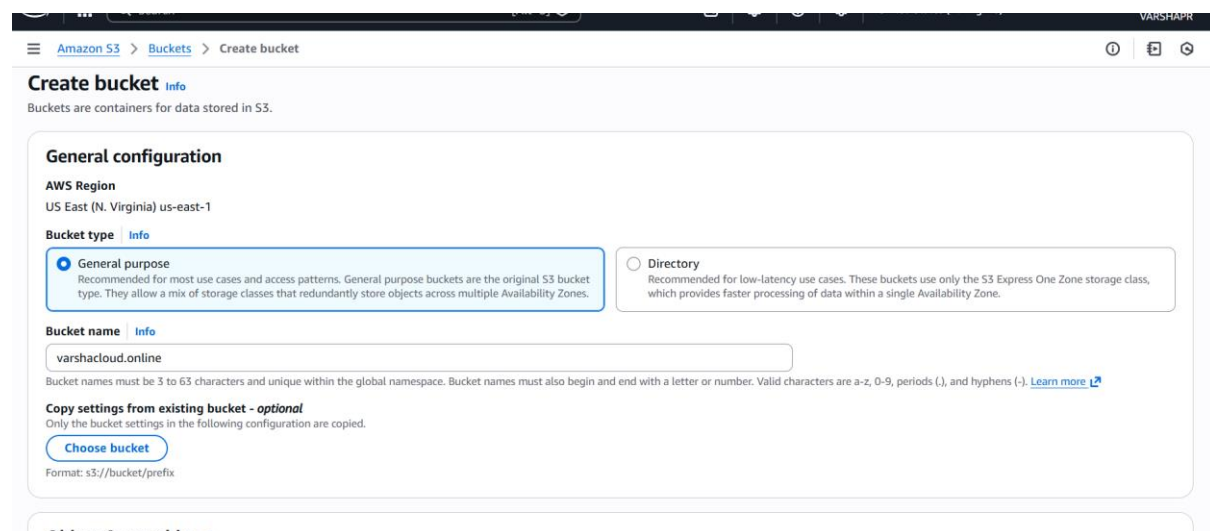
Your hosted zone is now created, and contains DNS records of type NS (Name Server) and SOA (Start of Authority).

Now you need to link your domain name purchased from a third-party domain registrar with your hosted zone. To do this, you will need to map the NS records of your hosted zone to the DNS configuration of the domain purchased from the domain registrar. The DNS configuration of the purchased domain can be easily found in the domain registrar's dashboard.

Create and Configure S3 Buckets:

To host your static website, you will need to create an S3 Bucket containing the pages of your website, then you will put the Bucket in public to make it accessible to all.

To begin with, create the main S3 Bucket which must have the same name as the website you are hosting:



The screenshot displays the AWS Management Console interface for creating a new S3 bucket. The breadcrumb navigation at the top shows 'Amazon S3 > Buckets > Create bucket'. The main heading is 'Create bucket' with an 'Info' link. Below this, a note states 'Buckets are containers for data stored in S3.' The 'General configuration' section is expanded, showing the 'AWS Region' as 'US East (N. Virginia) us-east-1'. Under 'Bucket type', the 'General purpose' option is selected, while 'Directory' is unselected. The 'Bucket name' field contains 'varshacloud.online'. A note below the name field specifies that bucket names must be 3 to 63 characters long and unique. At the bottom of the configuration section, there is an optional step 'Copy settings from existing bucket - optional' with a 'Choose bucket' button and a format example 's3://bucket/prefix'.

Then allow this Bucket to be publicly available on the Internet:

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Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

- ☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
- ☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
 - ☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
 - ☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
 - ☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Turning off block all public access might result in this bucket and the objects within becoming public
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

The main Bucket is now created, it is time to install the files of your website. To do this, go to your Bucket and click on Upload:

Amazon S3 > Buckets > varshacloud.online > Upload

Upload

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (0)

Remove

Add files

Add folder

All files and folders in this table will be uploaded.

Find by name

< 1 >

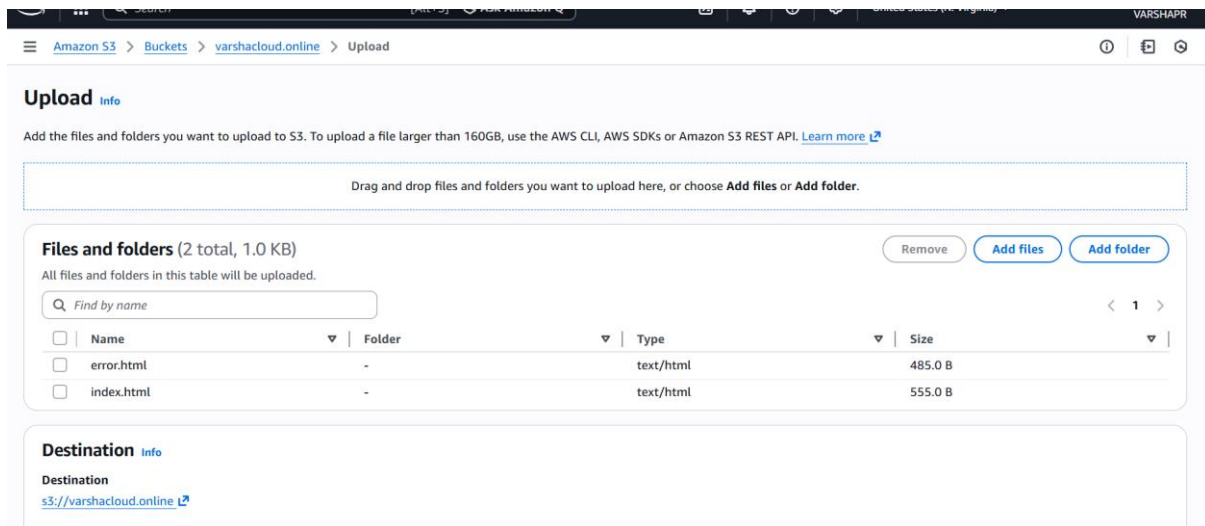
Name	Folder	Type	Size
No files or folders			
You have not chosen any files or folders to upload.			

Destination

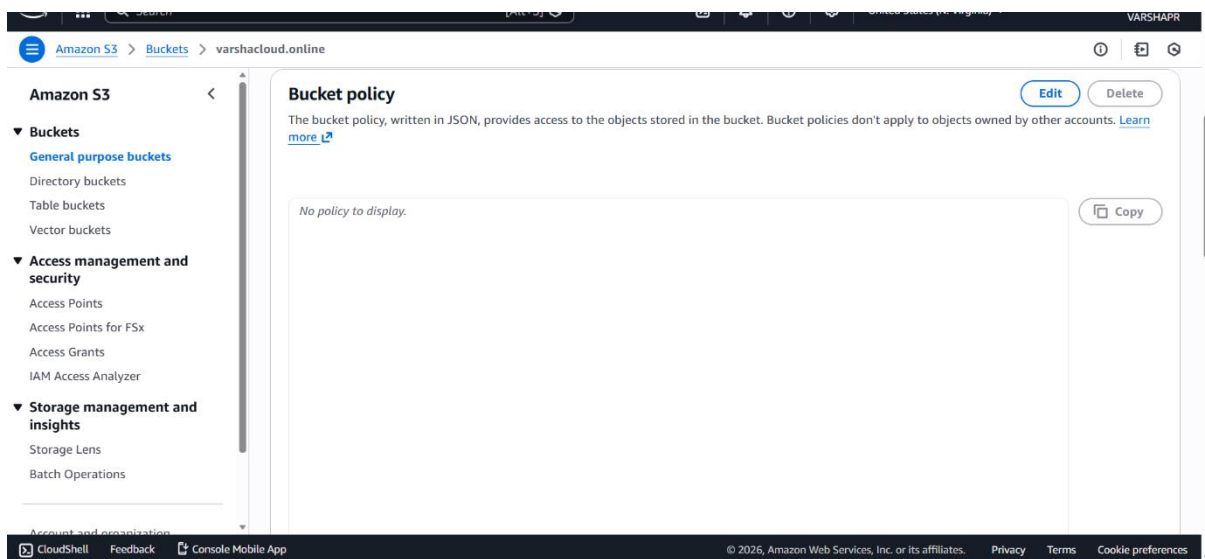
Destination

Add previously downloaded *index.html* and *error.html* files. The files are now objects in the main S3 Bucket:

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Move to **Permissions** to modify the Bucket Policy:



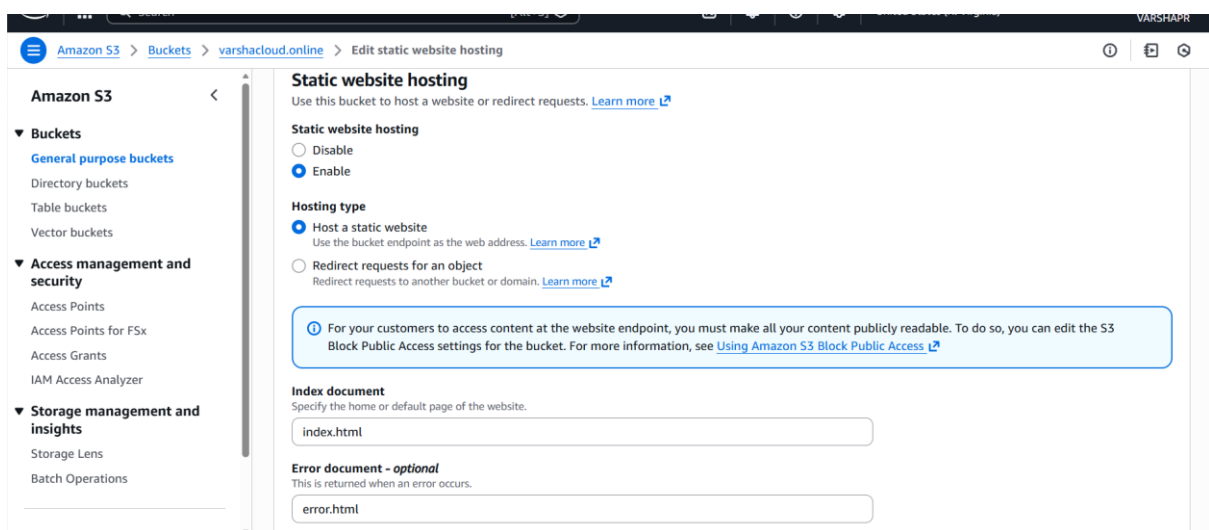
Insert the following policy and modify it slightly before validating:

```
{  
  "Version": "2012-10-17",  
  "Statement": [  
    {  
      "Sid": "PublicReadGetObject",
```

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```
"Effect": "Allow",
"Principal": "*",
"Action": [
    "s3:GetObject"
],
"Resource": [
    "arn:aws:s3:::bucket_name/*"
]
}
```

Last step: activate static web hosting by going to the **Static website hosting section**, at the bottom of the **Properties tab**:



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After saving changes, all you need to do is check that the site is accessible from a public address. Go back to the Static website hosting section to get the temporary URL of the S3 Bucket:

Now let's see what the S3 Bucket displays:

Welcome to VarshaCloud

This is my AWS static website hosted on S3 + CloudFront.

Explore my projects and portfolio!

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create the redirect S3 Bucket that will redirect all the traffic it receives to the main S3 Bucket. Then, go to the **Bucket configuration > Properties tab > Static website hosting section**:

activate the static website hosting, but this time by redirecting the requests. Specify the target Bucket , and leave the **Protocol field** as it is for the moment :

The screenshot shows the AWS Management Console interface for editing static website hosting on a bucket named 'www.varshacloud.online'. The breadcrumb navigation at the top reads: 'Amazon S3 > Buckets > www.varshacloud.online > Edit static website hosting'. The main heading is 'Edit static website hosting' with an 'Info' link. Below this, the 'Static website hosting' section includes a description and a 'Learn more' link. The 'Static website hosting' options are 'Disable' and 'Enable', with 'Enable' selected. The 'Hosting type' section has two options: 'Host a static website' and 'Redirect requests for an object', with the latter selected. The 'Host name' field contains 'varshacloud.online'. The 'Protocol - Optional' section has 'none' selected. The page is partially cut off at the bottom.

Amazon S3 > Buckets > www.varshacloud.online > Edit static website hosting

Edit static website hosting [Info](#)

Static website hosting
Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

☐ Disable
☒ Enable

Hosting type

☐ Host a static website
Use the bucket endpoint as the web address. [Learn more](#)

☒ Redirect requests for an object
Redirect requests to another bucket or domain. [Learn more](#)

Host name

Target bucket website address or personal domain

Protocol - Optional

☒ none
☐ http
☐ https

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Link the domain to S3 Buckets with Route 53:

You want to route traffic from your main web address :

The screenshot shows the 'Quick create record' interface in the AWS Route 53 console. The breadcrumb trail at the top indicates the path: Route 53 > Hosted zones > varshacloud.online > Create record. The form is titled 'Quick create record' and includes a 'Switch to wizard' link. It contains several sections: 'Record 1' with a 'Delete' button; 'Record name' set to 'subdomain' and 'varshacloud.online' with a note to keep it blank for the root domain; 'Record type' set to 'A - Routes traffic to an IPv4 address and some AWS resources'; 'Alias' toggle set to 'On'; 'Route traffic to' section with 'Alias to S3 website endpoint', 'US East (N. Virginia)', and 's3-website-us-east-1.amazonaws.com'; 'Routing policy' set to 'Simple routing'; and 'Evaluate target health' toggle set to 'Yes'. An 'Add another record' button is at the bottom right.

- If your main address is `www`, then add `www` to the record name. If your main address is non-`www`, leave the record name as it is;
- Select the Record type `A` to route the traffic to the main S3 Bucket;
- Activate the Alias to specify the S3 Bucket to which you want to route the traffic;
- Choose an Alias to S3 website endpoint in the region where the Bucket is located (here `us-east-1`), and select the Bucket that is automatically displayed in the selection;
- Routing policy will be simple routing;
- No need to evaluate the target health since the website is hosted on S3.

Once the first record is created, wait several minutes before checking if your website is accessible via the main web address:

You want to route the traffic of your redirect web address:

- If your redirect address is `www`, then add `www` to the record name. If your redirect address is non-`www`, leave the record name as it is;
- Select again a Record type in `A` to route the traffic to the redirect S3 Bucket;
- Activate the Alias to specify the S3 Bucket to which you want to route the traffic;
- Choose an Alias to S3 website endpoint in the region where the Bucket is located (here `us-east-1`), and select the Bucket that is automatically displayed in the selection;

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- Routing policy will be simple routing;
- No need to evaluate the target health since the site is hosted on S3.

Create SSL Certificate with AWS Certificate Manager:

To create a SSL certificate for your website, first go to AWS Certificate Manager and request a certificate. The certificate will be used on a public website, so you need to **request a public certificate**:

Request certificate

Certificate type [Info](#)

ACM certificates can be used to establish secure communications access across the internet or within an internal network. Choose the type of certificate for ACM to provide.

- ☒ **Request a public certificate**
Request a public SSL/TLS certificate from Amazon. By default, public certificates are trusted by browsers and operating systems.
- ☐ **Request a private certificate**
No private CAs available for issuance.

Requesting a private certificate requires the creation of a private certificate authority (CA). To create a private CA, visit [AWS Private Certificate Authority](#) [↗](#)

[Cancel](#)

[Next](#)

After that, all you have to do is validate the SSL Certificate creation request:

Domain names

Provide one or more domain names for your certificate.

Fully qualified domain name [Info](#)

varshacloud.online

[Remove](#)

www.varshacloud.online

[Remove](#)

[Add another name to this certificate](#)

You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" so that customers can reach your site by either name.

Allow export [Info](#)

- ☒ **Disable export**
Use this certificate only with integrated AWS services. The private key for this certificate will be disallowed for exporting from AWS.

- ☐ **Enable export**
Export this certificate and private key for use with any TLS workflow. ACM will charge your account based on the requested domains when the certificate is issued for the first time and for each renewal.

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Validation method [Info](#)

Select a method for validating domain ownership.

☒ **DNS validation - recommended**

Choose this option if you are authorized to modify the DNS configuration for the domains in your certificate request.

☐ **Email validation**

Choose this option if you do not have permission or cannot obtain permission to modify the DNS configuration for the domains in your certificate request.

Key algorithm [Info](#)

Select an encryption algorithm. Some algorithms may not be supported by all AWS services.

☒ **RSA 2048**

RSA is the most widely used key type.

☐ **ECDSA P 256**

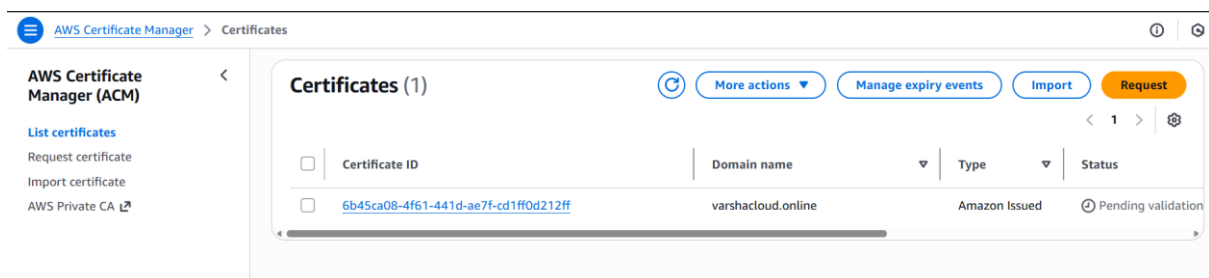
Equivalent in cryptographic strength to RSA 3072.

☐ **ECDSA P 384**

Equivalent in cryptographic strength to RSA 7680.

Tags [Info](#)

The certificate is created, but not yet validated; let's check its configuration:



In the configuration of the certificate, you need to add two DNS records of type CNAME in your hosted zone. Lucky you, AWS takes care of this: all you have to do is, first, select **Create records in Route 53**:

Then select the two FQDNs and finally **Create records**:

All you have to do now is wait a few minutes until you see the **Success status**:

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AWS Certificate Manager (ACM)

List certificates
Request certificate
Import certificate
AWS Private CA

Certificate status

Identifier
6b45ca08-4f61-441d-ae7f-cd1ff0d212ff

Status
Issued

ARN
arn:aws:acm:us-east-1:794694424957:certificate/6b45ca08-4f61-441d-ae7f-cd1ff0d212ff

Type
Amazon Issued

Domains (2)

Create records in Route 53 Export to CSV

Domain	Status	Renewal status	Type	CNAME name
varshacloud.online	Success	-	CNAME	_577f833215f80438001bne.
www.varshacloud.online	Success	-	CNAME	_a7b8fb0271928ef7a1b0online.

It's done, AWS has filled in the CNAME records for you. You can verify it by moving to your hosted zone:

☐	Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...
☐	varshaclo...	A	Simple	-	Yes	s3-website-us-east-1.amazo...	-	-
☐	varshaclo...	NS	Simple	-	No	ns-332.awsdns-41.com. ns-581.awsdns-08.net. ns-1102.awsdns-09.org. ns-1970.awsdns-54.co.uk.	172800	-
☐	varshaclo...	SOA	Simple	-	No	ns-332.awsdns-41.com. awsd...	900	-
☐	_577f833...	CNAME	Simple	-	No	_ac3aca27c9cfa92e06a1c6b...	300	-
☐	www.vars...	A	Simple	-	Yes	s3-website-us-east-1.amazo...	-	-
☐	_a7b8fb0...	CNAME	Simple	-	No	_42c6bf9bec4114ff3920efab...	300	-

Create and Configure Cloud Distributions:

By using CloudFront, you will be able to deploy your website in HTTPS, and get other benefits as using Edge locations to provide faster access to the website for the users.

Before creating your main CloudFront distribution, go to the main S3 Bucket, and copy the Bucket URL into **Properties tab, static website hosting section**:

Now, create your CloudFront distribution. Paste the Bucket URL you previously copied in Domain field:

Main CloudFront Distribution:

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CloudFront > Distributions > Create distribution

Step 1

Get started

Step 2

Specify origin

Step 3

Enable security

Step 4

Get TLS certificate

Step 5

Review and create

Review and create

General configuration

Distribution name

varshacloud.online.s3-website-us-east-1.amazonaws.com

Description

-

Domains to serve

varshacloud.online

Billing

Pay-as-you-go (\$0/month)

Edit

Origin

S3 origin

varshacloud.online.s3-website-us-east-1.amazonaws.com

Origin path

-

Enable Origin Shield

No

Connection attempts

3

Connection timeout

10

Edit

Cache settings

Viewer protocol policy

Redirect HTTP to HTTPS

Allowed HTTP methods

GET, HEAD

Cache policy

[CachingOptimized](#)

Origin request policy

-

Response headers policy

-

Edit

Security

Security protections

Enabled

Use monitor mode

No

Use existing WAF configuration

No

Edit

TLS certificate

ARN

[arn:aws:acm:us-east-1:794694424957:certificate/6b45ca08-4f61-441d-ae7f-cd1ff0d212ff](#)

Covered domains

varshacloud.online
www.varshacloud.online

Source

Amazon

Edit

Redirect CloudFront Distribution:

CloudFront > Distributions > Create distribution

Step 1

Get started

Step 2

Specify origin

Step 3

Enable security

Step 4

Get TLS certificate

Step 5

Review and create

Review and create

General configuration

Distribution name

www.varshacloud.online.s3-website-us-east-1.amazonaws.com

Description

-

Domains to serve

www.varshacloud.online

Billing

Pay-as-you-go (\$0/month)

Edit

Origin

S3 origin

www.varshacloud.online.s3-website-us-east-1.amazonaws.com

Origin path

-

Enable Origin Shield

No

Connection attempts

3

Connection timeout

10

Edit

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CloudFront > Distributions > Create distribution

Cache settings

Viewer protocol policy

Redirect HTTP to HTTPS

Allowed HTTP methods

GET, HEAD

Cache policy

[CachingOptimized](#)

Origin request policy

-

Response headers policy

-

Edit

Security

Security protections

Enabled

Use monitor mode

No

Use existing WAF configuration

No

Edit

TLS certificate

ARN

[arn:aws:acm:us-east-1:794694424957:certificate/6b45ca08-4f61-441d-ae7f-cd1ff0d212ff](#)

Covered domains

varshacloud.online
www.varshacloud.online

Source

Amazon

Edit

Go back to CloudFront, where you copy the web addresses of your CloudFront distributions (**Domain name field**) in the navigator search bar, in order to verify that you can access your site through these web addresses:

Distributions (2) Info

Enable

Disable

Delete

Search all distributions

Filter type
All distributions

ID	Status	Description	Type	Domain n...	Alternate ...	Origins	Pricing plan
E1220HMB5896WZ	Enabled	-	Standard	d2lk07e...	www.varshacloud.c	www.varshacloud.c	Pay-as-you-go
E1GJHRR97HAZ1K	Enabled	-	Standard	d2trxym...	varshacloud.online	varshacloud.online	Pay-as-you-go

Start by copying the URL of the redirect CloudFront distribution:

<https://d2trxymb1f5lw8.cloudfront.net>

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This is my AWS static website hosted on S3 + CloudFront.

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Check with Main CloudFront Distribution.

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To finalize, head to Route 53 to make changes to the A records, so that the web addresses route traffic to your freshly created Cloudfront distributions.

The record for the main domain name should look like this:

Edit record

 | 

Record name | [Info](#)

varshacloud.online

Keep blank to create a record for the root domain.

Record type | [Info](#)

A – Routes traffic to an IPv4 address and s... ▼

☒ Alias

Route traffic to | [Info](#)

Alias to CloudFront distribution ▼

US East (N. Virginia) ▼

An alias to a CloudFront distribution and another record in the same hosted zone are global and available only in US East (N. Virginia).

✕

Routing policy | [Info](#)

Simple routing ▼

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And the record for the redirect domain name should look like that:

Record name [Info](#)

.varshacloud.online

Keep blank to create a record for the root domain.

Record type [Info](#)

A – Routes traffic to an IPv4 address and s... ▼

☒ Alias

Route traffic to [Info](#)

Alias to CloudFront distribution ▼

US East (N. Virginia) ▼

An alias to a CloudFront distribution and another record in the same hosted zone are global and available only in US East (N. Virginia).

🔍 d2lk07evfokipz.cloudfront.net. ✕

Routing policy [Info](#)

Simple routing ▼

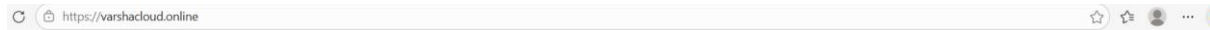
Evaluate target health

☐ No

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Let's check out the main and redirect domain in HTTP

(<http://varshacloud.online>)(<http://www.varshacloud.online>):



Welcome to VarshaCloud

This is my AWS static website hosted on S3 + CloudFront.

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<https://varshacloud.online>

Welcome to VarshaCloud

This is my AWS static website hosted on S3 + CloudFront.

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By following this step we will host a secure and scalable website on aws.

Outcome

- Successfully deployed a **secure, globally accessible static website** using AWS services.
- The website is hosted on **Amazon S3** with a custom domain (varshacloud.online).
- **Route 53** DNS records correctly link the domain to the hosting bucket.
- **SSL/TLS certificate** from AWS Certificate Manager ensures encrypted HTTPS communication.
- **CloudFront distribution** delivers content with low latency and optimized performance worldwide.
- Demonstrated practical skills in:
 - Cloud infrastructure setup (S3, Route 53, CloudFront, ACM).
 - Domain integration and DNS management.

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- Security implementation with SSL/TLS.
- Performance optimization using CDN caching.

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